

## Complete PHP Source Code

### Config

```
<?php
```

```
/**
```

```
 * Database connection settings.
```

```
 * Update the credentials below to match your local PostgreSQL setup.
```

```
 */
```

```
$DB_HOST = 'localhost';
```

```
$DB_PORT = '5432';
```

```
$DB_NAME = 'lab_login';
```

```
$DB_USER = 'postgres';
```

```
$DB_PASS = 'krys2002';
```

```
try {
```

```
    $pdo = new PDO(
```

```
        "pgsql:host={$DB_HOST};port={$DB_PORT};dbname={$DB_NAME}",
```

```
        $DB_USER,
```

```
        $DB_PASS,
```

```
        [
```

```
            PDO::ATTR_ERRMODE          => PDO::ERRMODE_EXCEPTION,
```

```
            PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC,
```

```
            PDO::ATTR_EMULATE_PREPARES => false,
```

```
        ]
```

```
    );
```

```
} catch (PDOException $e) {
```

```
    die('Database connection failed: ' . $e->getMessage());
```

```
}
```

### Public

#### INDEX.PHP

```
<?php
```

```
session_start();
```

```
require_once __DIR__ . '/../config/database.php';
```

```

// If already logged in, go straight to the dashboard
if (!empty($_SESSION['username'])) {
    header('Location: dashboard.php');
    exit;
}

$error = "";

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $username = trim($_POST['username'] ?? "");
    $password = $_POST['password'] ?? "";

    if ($username === "" || $password === "") {
        $error = 'Please enter both username and password.';
    } else {
        $stmt = $pdo->prepare("SELECT id, TRIM(username) AS username, password, fullname FROM users
WHERE LOWER(TRIM(username)) = LOWER(?) LIMIT 1");
        $stmt->execute([$username]);
        $user = $stmt->fetch();
        $storedPassword = $user['password'] ?? "";
        $passwordMatches = false;

        if ($user) {
            if (password_verify($password, $storedPassword)) {
                $passwordMatches = true;
            } elseif (hash_equals($storedPassword, $password)) {
                $passwordMatches = true;

                $newHash = password_hash($password, PASSWORD_DEFAULT);
                $update = $pdo->prepare("UPDATE users SET password = ? WHERE id = ?");
                $update->execute([$newHash, $user['id']]);
            }
        }
    }
}

```

```

    if ($passwordMatches) {
        // Regenerate session ID to prevent session fixation
        session_regenerate_id(true);
        $_SESSION['user_id'] = $user['id'];
        $_SESSION['username'] = $user['username'];
        $_SESSION['fullname'] = $user['fullname'] ?? $user['username'];

        header('Location: dashboard.php');
        exit;
    } else {
        $error = 'Invalid username or password.';
    }
}
?>

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Login System</title>
<link rel="stylesheet" href="assets/style.css">
</head>
<body>
<main class="auth-shell">
<section class="card auth-card">
    <div class="card-header">
        <p class="eyebrow">Welcome back</p>
        <h1>Login System</h1>

    </div>

    <?php if ($error): ?>

```

```

        <div class="error-message"><?= htmlspecialchars($error) ?></div>
    <?php endif; ?>

    <form method="POST" action="index.php">
        <div class="form-group">
            <label for="username">Username</label>
            <input type="text" id="username" name="username" autocomplete="username" required>
        </div>

        <div class="form-group">
            <label for="password">Password</label>
            <input type="password" id="password" name="password" autocomplete="current-password"
required>
        </div>

        <div class="btn-group">
            <button type="submit" name="login" class="btn-login">Login</button>
            <button type="reset" class="btn-reset">Reset</button>
        </div>
    </form>
</section>
</main>
</body>
</html>

```

## DASHBOARD.PHP

```

<?php
session_start();

// Protect this page: must be logged in
if (empty($_SESSION['username'])) {
    header("Location: index.php");
    exit;
}
?>

```

```

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Dashboard | Login System</title>
<link rel="stylesheet" href="assets/style.css">
</head>
<body>
<main class="auth-shell">
<section class="card dashboard-card">
    <div class="dashboard-badge">Active Session</div>
    <h1>Welcome, <?= htmlspecialchars($_SESSION['fullname']) ?? $_SESSION['username'] ?></h1>
    <p class="subtitle">You are signed in to the Simple Login System.</p>

    <div class="dashboard-panel">
        <span>Course</span>
        <strong>System Integration and Architecture 1</strong>
    </div>

    <a href="logout.php" class="btn-logout">Logout</a>
</section>
</main>
</body>
</html>

```

## LOGOUT.PHP

```

<?php
session_start();

// Clear all session variables
$_SESSION = [];

// Delete the session cookie, if one is used

```

```
if (ini_get('session.use_cookies')) {  
    $params = session_get_cookie_params();  
    setcookie(  
        session_name(),  
        "",  
        time() - 42000,  
        $params['path'],  
        $params['domain'],  
        $params['secure'],  
        $params['httponly']  
    );  
}
```

```
// Destroy the session on the server  
session_destroy();
```

```
// Redirect back to the login page  
header('Location: index.php');  
exit;  
CSS
```

```
* {  
    box-sizing: border-box;  
    margin: 0;  
    padding: 0;  
}
```

```
:root {  
    --bg-start: #133561;  
    --bg-end: #eefdf5;  
    --surface: #ffffff;  
    --surface-soft: #f8fafc;  
    --text: #172033;  
    --muted: #64748b;  
    --line: #dbe4ef;
```

```
--primary: #2563eb;
--primary-dark: #1d4ed8;
--success: #16a34a;
--danger: #dc2626;
--danger-dark: #b91c1c;
--shadow: 0 24px 70px rgba(30, 41, 59, 0.14);
}
```

```
body {
  min-height: 100vh;
  color: var(--text);
  font-family: "Segoe UI", Arial, sans-serif;
  background:
    radial-gradient(circle at top left, rgba(37, 99, 235, 0.18), transparent 32rem),
    radial-gradient(circle at bottom right, rgba(22, 163, 74, 0.14), transparent 28rem),
    linear-gradient(135deg, var(--bg-start), var(--bg-end));
}
```

```
.auth-shell {
  min-height: 100vh;
  display: flex;
  align-items: center;
  justify-content: center;
  padding: 32px 18px;
}
```

```
.card {
  width: 100%;
  max-width: 420px;
  padding: 34px;
  border: 1px solid rgba(219, 228, 239, 0.9);
  border-radius: 8px;
  background: rgba(255, 255, 255, 0.92);
  box-shadow: var(--shadow);
}
```

```
}
```

```
.card-header {  
  margin-bottom: 26px;  
  text-align: center;  
}
```

```
.eyebrow {  
  margin-bottom: 8px;  
  color: var(--primary);  
  font-size: 0.78rem;  
  font-weight: 800;  
  letter-spacing: 0.08em;  
  text-transform: uppercase;  
}
```

```
.card h1,  
.dashboard-card h1 {  
  color: var(--text);  
  font-size: 1.8rem;  
  line-height: 1.2;  
}
```

```
.subtitle {  
  margin-top: 8px;  
  color: var(--muted);  
  font-size: 0.98rem;  
  line-height: 1.5;  
}
```

```
.form-group {  
  margin-bottom: 18px;  
}
```

```
.form-group label {  
  display: block;  
  margin-bottom: 8px;  
  color: #334155;  
  font-size: 0.92rem;  
  font-weight: 700;  
}
```

```
.form-group input[type="text"],  
.form-group input[type="password"] {  
  width: 100%;  
  padding: 12px 14px;  
  border: 1px solid var(--line);  
  border-radius: 8px;  
  background: var(--surface-soft);  
  color: var(--text);  
  font-size: 1rem;  
  transition: border-color 0.2s ease, box-shadow 0.2s ease, background 0.2s ease;  
}
```

```
.form-group input:focus {  
  outline: none;  
  border-color: var(--primary);  
  background: var(--surface);  
  box-shadow: 0 0 0 4px rgba(37, 99, 235, 0.14);  
}
```

```
.btn-group {  
  display: grid;  
  grid-template-columns: 1fr 1fr;  
  gap: 12px;  
  margin-top: 24px;  
}
```

```
button,  
.btn-logout {  
    min-height: 46px;  
    border: 0;  
    border-radius: 8px;  
    font-size: 1rem;  
    font-weight: 800;  
    cursor: pointer;  
    transition: transform 0.2s ease, box-shadow 0.2s ease, background 0.2s ease;  
}
```

```
button:hover,  
.btn-logout:hover {  
    transform: translateY(-1px);  
}
```

```
.btn-login {  
    background: var(--primary);  
    color: #fff;  
    box-shadow: 0 12px 24px rgba(37, 99, 235, 0.24);  
}
```

```
.btn-login:hover {  
    background: var(--primary-dark);  
}
```

```
.btn-reset {  
    border: 1px solid var(--line);  
    background: #fff;  
    color: #475569;  
}
```

```
.btn-reset:hover {  
    background: #f1f5f9;
```

```
}
```

```
.error-message {  
  margin-bottom: 18px;  
  padding: 12px 14px;  
  border: 1px solid #fecaca;  
  border-radius: 8px;  
  background: #fef2f2;  
  color: #991b1b;  
  font-size: 0.92rem;  
  line-height: 1.45;  
  text-align: center;  
}
```

```
.dashboard-card {  
  max-width: 520px;  
  text-align: center;  
}
```

```
.dashboard-badge {  
  display: inline-flex;  
  align-items: center;  
  justify-content: center;  
  min-height: 34px;  
  margin-bottom: 18px;  
  padding: 7px 14px;  
  border-radius: 999px;  
  background: #dcfce7;  
  color: #166534;  
  font-size: 0.82rem;  
  font-weight: 800;  
}
```

```
.dashboard-panel {
```

```
display: grid;
gap: 6px;
margin: 26px 0;
padding: 18px;
border: 1px solid var(--line);
border-radius: 8px;
background: var(--surface-soft);
text-align: left;
}
```

```
.dashboard-panel span {
  color: var(--muted);
  font-size: 0.82rem;
  font-weight: 800;
  text-transform: uppercase;
}
```

```
.dashboard-panel strong {
  color: var(--text);
  font-size: 1rem;
  line-height: 1.4;
}
```

```
.btn-logout {
  display: inline-flex;
  align-items: center;
  justify-content: center;
  min-width: 150px;
  padding: 0 24px;
  background: var(--danger);
  color: #fff;
  text-decoration: none;
  box-shadow: 0 12px 24px rgba(220, 38, 38, 0.2);
}
```

```
.btn-logout:hover {  
  background: var(--danger-dark);  
}
```

```
@media (max-width: 480px) {  
  .auth-shell {  
    padding: 20px 14px;  
  }
```

```
.card {  
  padding: 26px 20px;  
}
```

```
.card h1,  
.dashboard-card h1 {  
  font-size: 1.55rem;  
}
```

```
.btn-group {  
  grid-template-columns: 1fr;  
}  
}
```

SQL

-- Create table

```
CREATE TABLE IF NOT EXISTS users (  
  id SERIAL PRIMARY KEY,  
  username VARCHAR(50) UNIQUE NOT NULL,  
  password VARCHAR(255) NOT NULL,  
  fullname VARCHAR(100) NOT NULL  
);
```

```
INSERT INTO users (username, password, fullname) VALUES  
('Mark', '$2y$12$NiL6EP3zisPwat4ejkiEXeHczQhZq.IVYdmntTnQpSLxOEiYVXk8y', 'Mark'),
```

```
('Kyle', '$2y$12$n362nLPpYZ1m.ZAMfsOTT.1BTyAfAL8rYdSxcjDn0S48QeC9VG5Am', 'Kyle'),  
('Kram', '$2y$12$OasX6PzoTDS7zolrAWohd.Ax7DeNvoKbFjKuPqokr8JcWzM2bLUgG', 'Kram')
```

```
ON CONFLICT (username) DO UPDATE SET
```

```
password = EXCLUDED.password,
```

```
fullname = EXCLUDED.fullname;
```